Semantic Plagiarism Detection Using Embedding Similarity

Motivation

It's hard to detect whether students cheat on a writing assignment at scale. I study whether embedding similarity separates plagiarism-like pairs, formed by either swapping synonyms, sentences, or LLM rewriting, from essays that share only the same assignment topic.

Experimental Setup

- Dataset: 450 essays from PERSUADE 2.0, sampled as 15 assignment groups x 30 essays.
- Positive pairs: 5 generated plagiarism variants for each essay.
- Negatives: 3 same-assignment soft negatives and 1 different-assignment hard negative per essay.
- Scale: 2,250 plagiarism pairs, 1,350 soft negatives, and 450 hard negatives.

Methodology

Sample 450 essays 15 prompts → **Generate** 5 plagiarism variants → **Encode** 9 embedding models → **Compare** cosine + other metrics → **Decide** train threshold holdout F1

Pairs are encoded as two vectors and compared by similarity. The final baseline freezes the encoder, calibrates one cosine threshold on train, and applies it to grouped holdout essays.

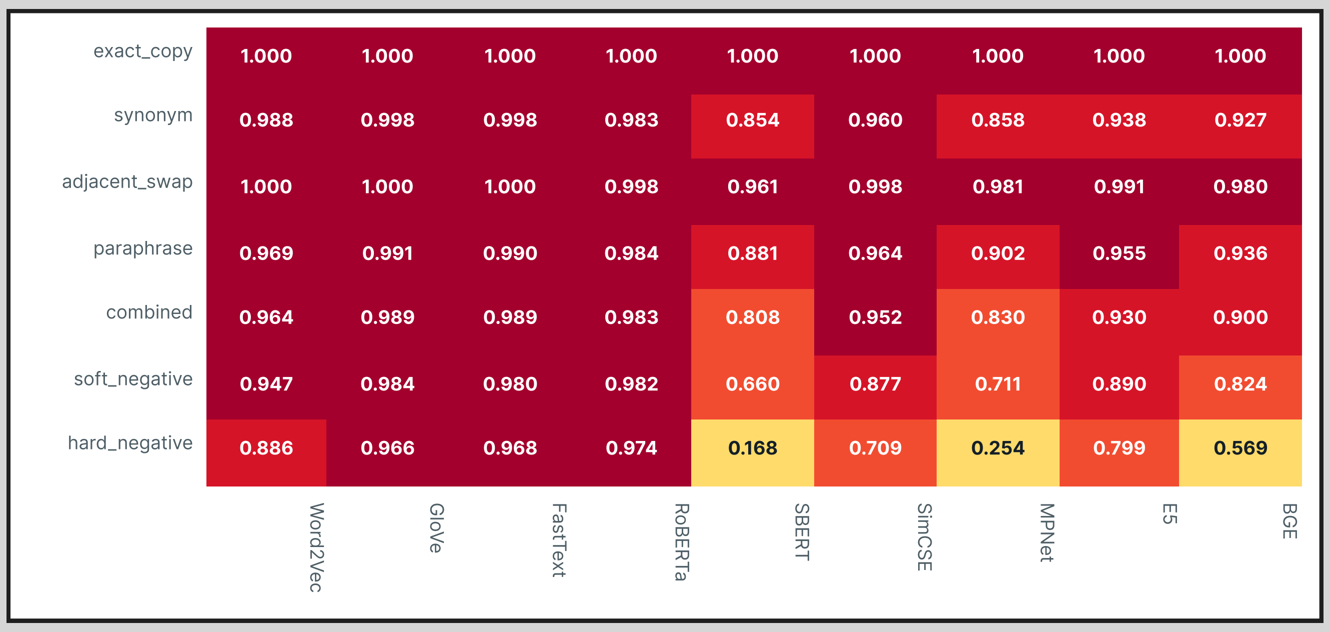
Models And Metrics

- Static vectors: Word2Vec, GloVe, FastText with mean pooling.
- Contextual baseline: RoBERTa with simple mean pooling.
- Similarity-trained encoders: SBERT, SimCSE, Paraphrase-MPNet, E5, BGE.
- Metrics: separation score → ROC-AUC → train-calibrated F1.

Conclusions

- Cosine similarity is the cleanest readout for sentence-embedding models.
- SimCSE is strongest under the grouped holdout cosine-threshold baseline.
- Soft negatives make the task realistic: the model must separate same topic from same meaning.
- The concatenated linear probe is not useful as a tool for reliably contrast texts, unlike direct embedding similarity.

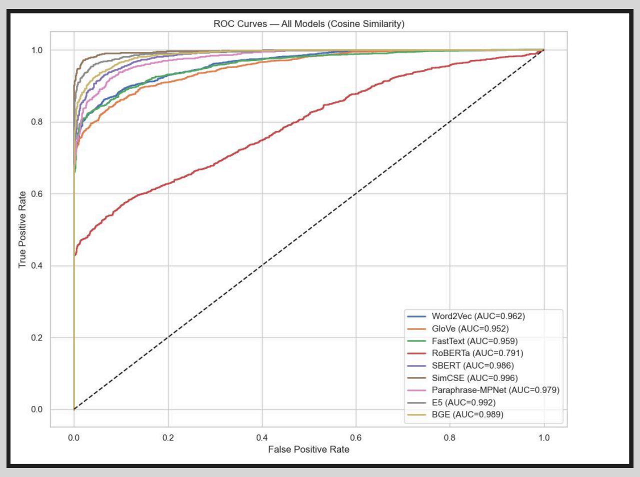
Results: Similarity Geometry



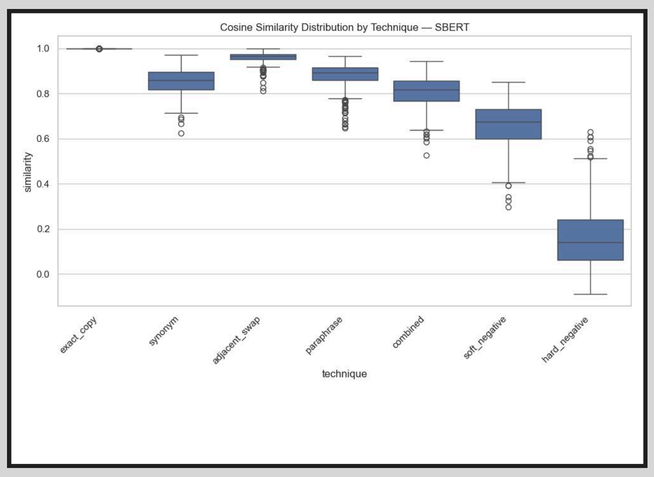
Mean cosine similarity by plagiarism technique and model.

Soft negatives test whether an encoder separates topic similarity from semantic equivalence. Static vector models having hard time differentiating even hard negatives.

ROC And Score Separation



ROC curves.

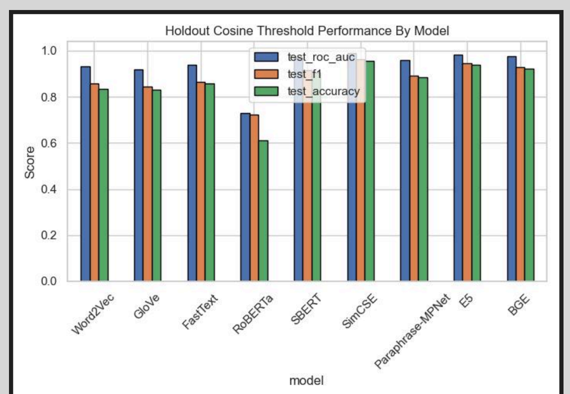


Score distributions.

SBERT has the largest same-assignment margin (0.2404). SimCSE has the strongest cosine ROC-AUC (0.9962).

Holdout Cosine Threshold Baseline

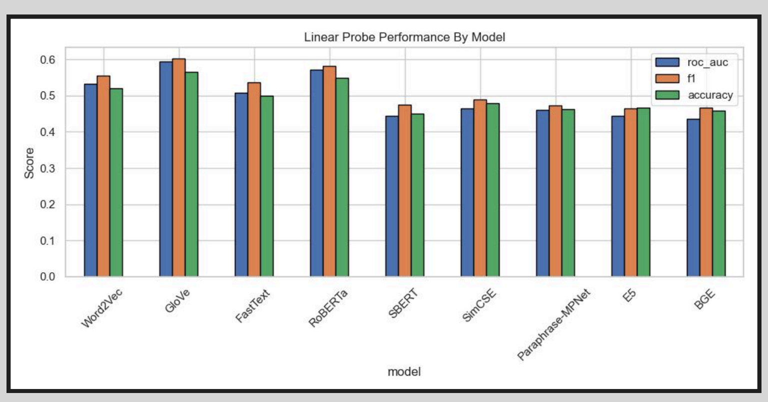
model	thr.	ROC-AUC	F1	acc.
SimCSE	0.9347	**0.9891**	**0.9613**	0.9572
E5	0.9178	**0.9819**	**0.9441**	0.9383
BGE	0.8837	**0.9765**	**0.9279**	0.9202
SBERT	0.7769	0.9702	0.9141	0.9045
MPNet	0.8202	0.9585	0.8920	0.8840
FastText	0.9905	0.9403	0.8638	0.8568
Word2Vec	0.9612	0.9307	0.8561	0.8329
GloVe	0.9897	0.9197	0.8452	0.8296
RoBERTa	0.9760	0.7284	0.7228	0.6099



Holdout threshold results.

Thresholds are calibrated on train and frozen for holdout, best order: SimCSE, E5, BGE, then SBERT.

Linear Probe Contrast



A linear classifier over two sentences makes GloVe/ RoBERTa look stronger, suggesting lexical or prompt-correlated cues.